DIY Wireless Sim Racing Button Box

Component Cost Breakdown • GitHub Project Documentation

This document summarizes the known component costs and estimated PETG 3D-printing cost for the wireless sim racing steering-wheel button box. USD conversions are estimates based on **1 USD ≈ 280 PKR**.

1. Electronics & Components

Component	Qty.	Unit Price (PKR)	Total (PKR)	Total (USD)
ESP32-WROOM-32	1	1,150	1,150	$4.11
Connecting wires	1 set	50	50	$0.18
Potentiometers	4	30	120	$0.43
Push buttons (momentary)	8	60	480	$1.71
Toggle switches	2	100	200	$0.71
Potentiometer knobs	4	30	120	$0.43
Lithium-ion cell (2000 mAh)	1	350	350	$1.25
Battery charging module	1	150	150	$0.54
Electronics subtotal	**—**	**—**	**2,620**	**$9.36**

2. PETG 3D-Printed Body

The enclosure is 3D-printed in PETG. Approximate filament consumption is **300 g**. The estimates below use the supplied per-gram costs for self-printing and a third-party printing service.

Printing method	Rate	Filament	Estimated total (PKR)	Estimated total (USD)
Self-printing	PKR 5/g	300 g	1,500	$5.36
Printing service	PKR 12/g	300 g	3,600	$12.86

3. Estimated Project Total

Cost scenario	Calculation	Total (PKR)	Total (USD)
Self-printed body	PKR 2,620 + PKR 1,500	**4,120**	**$14.71**
Printing service body	PKR 2,620 + PKR 3,600	**6,220**	**$22.21**

4. Notes & Exclusions

• USD values are approximate and calculated at 1 USD ≈ 280 PKR; exchange rates can change.

• Totals include the listed electronics, the 2000 mAh lithium-ion cell, its charging module, and estimated PETG body printing.

• The rear quick-release hardware and any other unlisted parts are not included because their prices have not been provided.

• Electricity, printer maintenance, failed prints, taxes, shipping, and assembly labor are not included unless covered by the stated per-gram rate.